

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

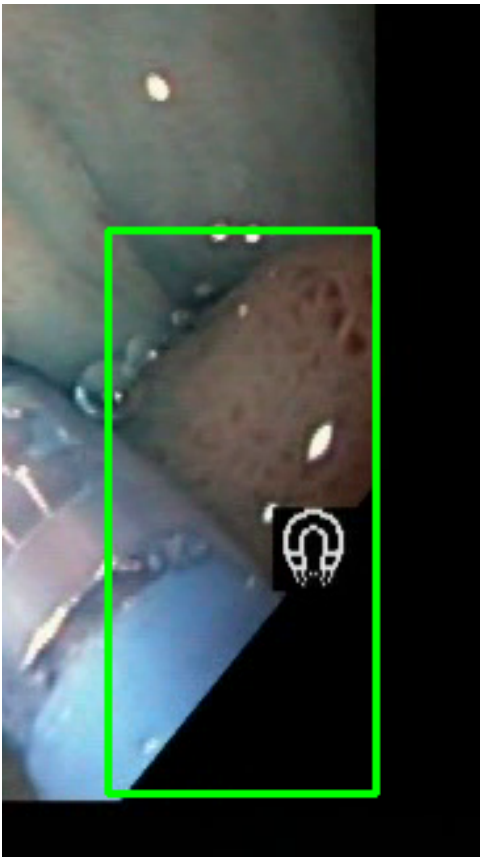
Video ID:	2330e8a9-3238-4e3f-8d77-0d6ff5dbc0f1
Total Frames:	8879
Frame Rate:	25.0 fps (assumed)
Duration:	355.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	4675	✓
RT-DETR Detections	6349	✓
MedSAM2 Detections	6293	✓
Consensus (3-Model Agreement)	4	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	3	75.0%
LOW RISK	1	25.0%
TOTAL ANALYZED	4	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 625 frames)



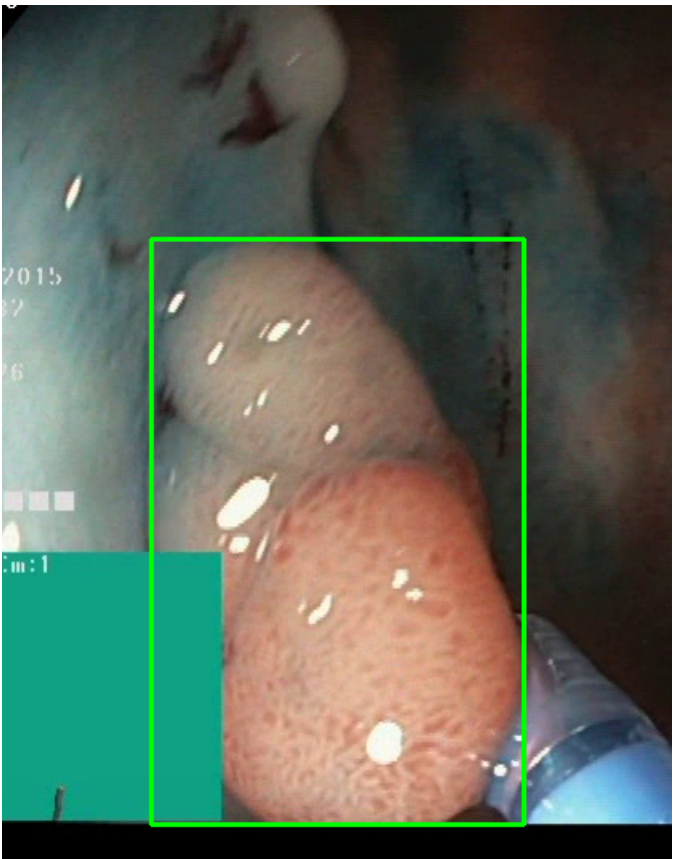
Feature	Value	Interpretation
Frame Index	2744	Frame 2744 of 8879
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 77.7%
Redness Score	0.028	Color-based vascularization
Relative Radius	27.3%	Polyp size as % of frame diagonal
Texture Score	0.538	Surface morphology
Vessel Visibility	0.008	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	88.3%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 2932 frames)



Feature	Value	Interpretation
Frame Index	2067	Frame 2067 of 8879
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 82.4%

Redness Score	0.061	Color-based vascularization
Relative Radius	24.2%	Polyp size as % of frame diagonal
Texture Score	0.117	Surface morphology
Vessel Visibility	0.338	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	89.7%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #3 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 66 frames)



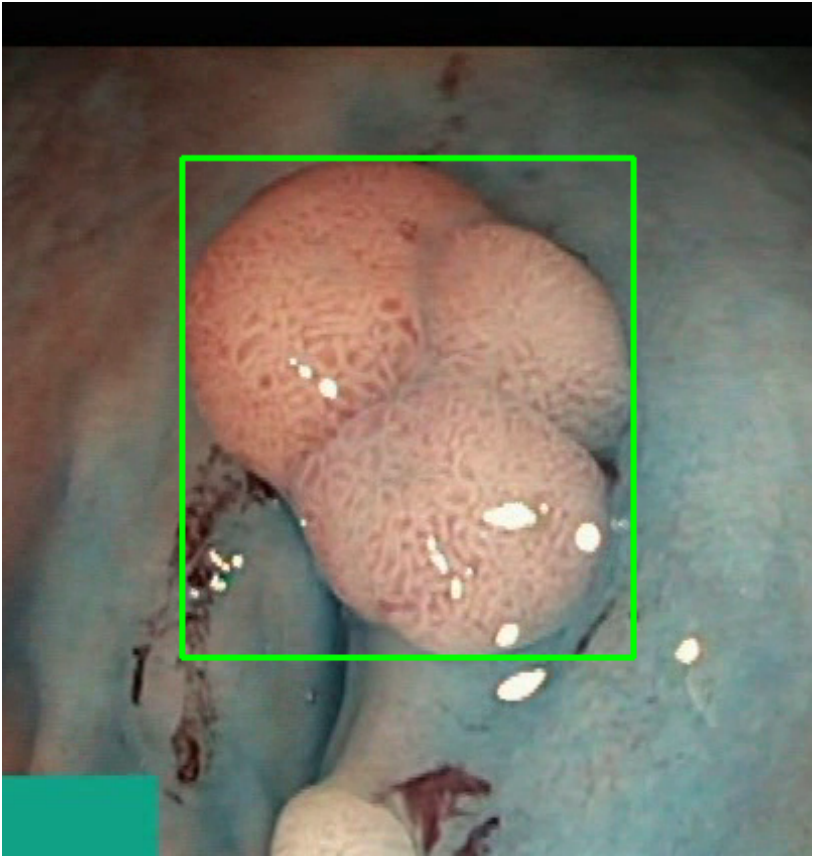
Feature	Value	Interpretation
Frame Index	3228	Frame 3228 of 8879
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 81.0%
Redness Score	0.079	Color-based vascularization
Relative Radius	25.4%	Polyp size as % of frame diagonal
Texture Score	0.087	Surface morphology

Vessel Visibility	0.441	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	84.7%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #4 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 237 frames)



Feature	Value	Interpretation
Frame Index	3433	Frame 3433 of 8879
Detection Method	Detector Consensus (MedSAM2) / Detector Agreement (MedSAM2 unavailable for this video)	
POLYP TYPE	FLAT_POLYP	Type confidence: 80.7%
Redness Score	0.083	Color-based vascularization
Relative Radius	25.0%	Polyp size as % of frame diagonal
Texture Score	0.120	Surface morphology
Vessel Visibility	0.394	Vascularization indicator
Risk Tier	MEDIUM_RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	88.0%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:

A total of 4 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 3 (75.0%)
- LOW RISK polyps: 1 (25.0%)

Recommendation:

The presence of MEDIUM RISK polyps warrants clinical review and correlation with endoscopic findings.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.